Journal of the Geological Society, London, Vol. **157**, 2000, pp. 1187–1197. Printed in Great Britain.

Large clockwise rotations in an extensional allochthon, Alboran Domain (southern Spain)

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Abstract: Palaeomagnetic, structural and geochronological analysis of a suite of Oligocene mafic dykes intruded into the Malaguide allochthon in the Internal Zone of the Betic Cordillera demonstrate that at least 134° clockwise vertical-axis rotation of a large region occurred, probably during early Miocene thrusting onto the External Zones. Initial results from Ar/Ar whole rock analysis suggests that the dykes were probably intruded over a period of time between 23 Ma and 30 Ma. Results of demagnetization experiments demonstrate that the remanence in many of the dykes is composed of one to three components. The lowest temperature component is poorly defined. The intermediate temperature (IT) component, carried by pyrrhotite, has reversed polarity, a declination of 314° and an inclination of −48°. This component was probably acquired during cooling after greenschist-facies metamorphism. The high temperature (HT) component is carried by magnetite and was probably acquired at or shortly after intrusion. This component has both reversed and normal polarity, with an average declination of 320° and an inclination of −13°. Rotation of approximately 35° about a horizontal NNE-trending axis, in present-day coordinates, brings the inclination of the HT component into statistical agreement with both the IT component and the expected inclination for the Oligocene. This rotation leaves the dykes in a near-vertical orientation but brings the foliation in the surrounding schists to approximately horizontal.

We suggest that the original regional foliation may have been sub-horizontal and related to the ductile phase of late-orogenic extension in the Betic Cordillera. The dykes were intruded sub-vertically into the schists with an original NW–SE trend. The region was then tilted and rotated clockwise by 23° ± 15 before acquiring the IT component after which it underwent 134° ± 10° of clockwise rotation about a vertical axis.

Keywords: Betic Cordillera, Oligocene, palaeomagnetism, dykes, absolute age.

The Internal Zone of the Betic Cordillera of southern Spain (Fig. 1) forms part of an Alpine collisional orogen, known as the Alboran Domain, that underwent late orogenic extensional collapse in the Miocene to form the present-day Alboran Sea in the Western Mediterranean (Platt & Vissers 1989; García-Dueñas *et al.* 1992). The deformational history involved (1) large-scale thrusting of Palaeozoic to Early Tertiary rocks, accompanied by high *P*/*T* ratio metamorphism, during Late Cretaceous? to Early Tertiary time, (2) ductile extension, accompanied in places by medium to low *P*/*T* ratio metamorphism during Late Oligocene to Early Miocene time (Vissers *et al*. 1995; Platt *et al*. 1998; Argles et al. 1999; Platt & Whitehouse 1999) and (3) brittle extension leading to the formation of marine sedimentary basins during the Miocene (Comas *et al*. 1999). The Alboran Domain is surrounded to north, west, and south by the External Betic–Rif arc, a Miocene thin-skinned fold and thrust belt that accommodated both the extension of the Alboran Domain and the continuing convergence of Africa and Iberia (Platt & Vissers 1989; García-Dueñas *et al*. 1992; Platzman 1992). Thrusting in the External Betic–Rif arc and in unmetamorphosed rocks of the Dorsale Complex along the margin of the Alboran Domain was accompanied by large clockwise vertical-axis rotations (Osete *et al.* 1988; Platzman 1992; Platzman & Lowrie 1992; Allerton *et al*. 1993; Platzman *et al.* 1993; Platt *et al.* 1995), which contributed to the development of the structural geometry at scales ranging from that of individual thrust sheets to that of the arcuate orogen as a whole (e.g., Kirker & Platt 1998).

Relatively little palaeomagnetic work has been carried out in the Internal Zone, mainly because of the widespread metamorphism, structural complexity, and the scarcity of rocks with well-developed magnetic remanences in this region. Results from the peridotite massifs in Spain (Fig. 1) and in Morocco, however, suggest that these rocks may have undergone about 45° and 74° of clockwise and anticlockwise rotations respectively (Saddiqi *et al.* 1995; Feinberg *et al.* 1996) although it was not possible to correct these results for the possible effects of later tilting. The sense and amount of these rotations are comparable to those found in the External Betic–Rif arc. Work on unmetamorphosed Mesozoic and Tertiary sediments much further east, in the Sierra Espuña (Fig. 1), suggest progressive clockwise rotation of about 180° during the late Oligocene to early Miocene (Allerton *et al*. 1993). Data from middle to late Miocene volcanic rocks in the Cabo de Gata region (Fig. 1) show evidence of both clockwise and anticlockwise rotation (Calvo *et al.* 1997), possibly associated with late Miocene to recent strike-slip faults in the area. It seems likely that vertical-axis rotations have played at least as important a role in the tectonic evolution of the Internal Betic–Rif arc as in the External arc. The purpose of this paper is to present an integrated palaeomagnetic, structural and geochronological study of part of a suite of Tertiary dykes within the Internal Zones of the Betic Cordillera, which sheds some light on the

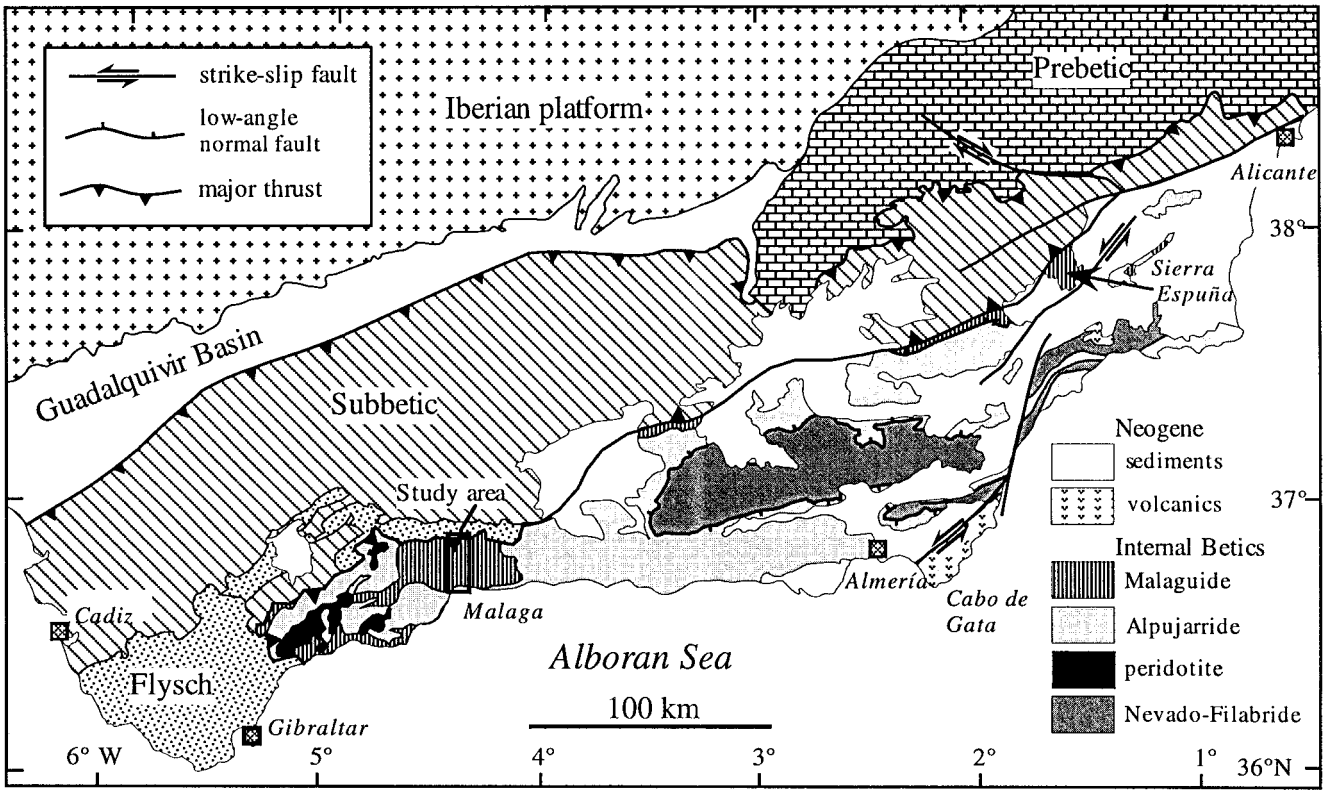


Fig. 1. Tectonic map of the Betic Cordilleras showing location of profile (box).

timing of vertical-axis rotations and their role in the tectonic evolution of the region.

Geological setting

The main body of the Internal Zone of the Betic Cordillera is traditionally divided into three major tectonic complexes: in descending order the Malaguide, Alpujarride, and Nevado–Filabride Complexes. These were originally regarded as nappe complexes derived from differing palaeogeographical domains (e.g., Egeler & Simon 1969), but the boundaries between them are now thought to be major low-angle normal sense faults or shear zones that have reactivated or cut earlier thrust contacts (Platt 1986; Platt & Vissers 1989; García-Dueñas *et al.* 1992; Jabaloy *et al.* 1993; Lonergan & Platt 1995). The timing of events in the Internal Zone is currently debated (compare Platt 1998 and Balanyá *et al.* 1997, for example), but radiometric data strongly suggest a period of rapid exhumation and cooling in early Miocene time, associated with the extensional phase of tectonics that created the Alboran Sea and related extensional basins onshore (Zeck *et al.* 1992; Monié *et al.*, 1994; Platt *et al.* 1998; Platt & Whitehouse 1999). The peripheral regions of the extended Internal Zone were emplaced onto the African and Iberian continental margins during the Miocene, to form the presently emergent Internal Zones of the Betic and Rif orogens. Extensional tectonics in the Internal Zone and thrusting in the External Betic–Rif arc appear to have overlapped in time.

The area we have studied lies within a region of approximately 50 km by 30 km to the north of the city of Malaga that has been assigned to the Malaguide Complex (Fig. 1). Three broadly defined geological units can be distinguished (Fig. 2). The lowest unit consists of low-grade phyllites and psammites of probable Early Palaeozoic age, locally carrying biotite and andalusite; this is overlain by a sequence of Late Palaeozoic carbonate and clastic rocks showing very low-grade metamorphism; and these are overlain in turn by apparently unmetamorphosed Permo-Triassic to Cretaceous clastic and carbonate rocks (IGME 1978, 1979; Tubia *et al.* 1993). This sequence is generally regarded as being stratigraphic in nature, and the contacts are regionally sub-horizontal. The contact between the middle and upper units is likely to be an unconformity of post-Variscan age. The nature of the contact between the lower and middle units is unclear, but in view of the contrast in metamorphic grade across it, it may be a late Alpine extensional fault. The lowest and middle units both show regionally developed tectonic fabrics, and are cut by numerous brittle normal faults. The structural geometry and history of the lower unit is discussed in more detail below.

The Malaguide rocks are cut by a suite of altered mafic dykes. These show relict magmatic textures, with primary zoned plagioclase, clinopyroxene and/or hornblende, and ilmenite. The primary grain size varies from 0.2 mm to 1 mm, and some dykes contain 5 mm plagioclase phenocrysts, now heavily altered. Plagioclase is altered to varying degrees to white mica and epidote minerals, and the mafic phases are largely altered to fine grained feathery aggregates of pale green clino-amphibole. Ilmenite is now largely preserved as skeletal crystals in which exsolved magnetite lamellae have been altered, and the ilmenite itself partly altered to sphene. Secondary titanite, chlorite, white mica and pyrrhotite are abundant, and some samples contain metamorphic biotite. There is no textural evidence for ductile deformation. Most of the dykes

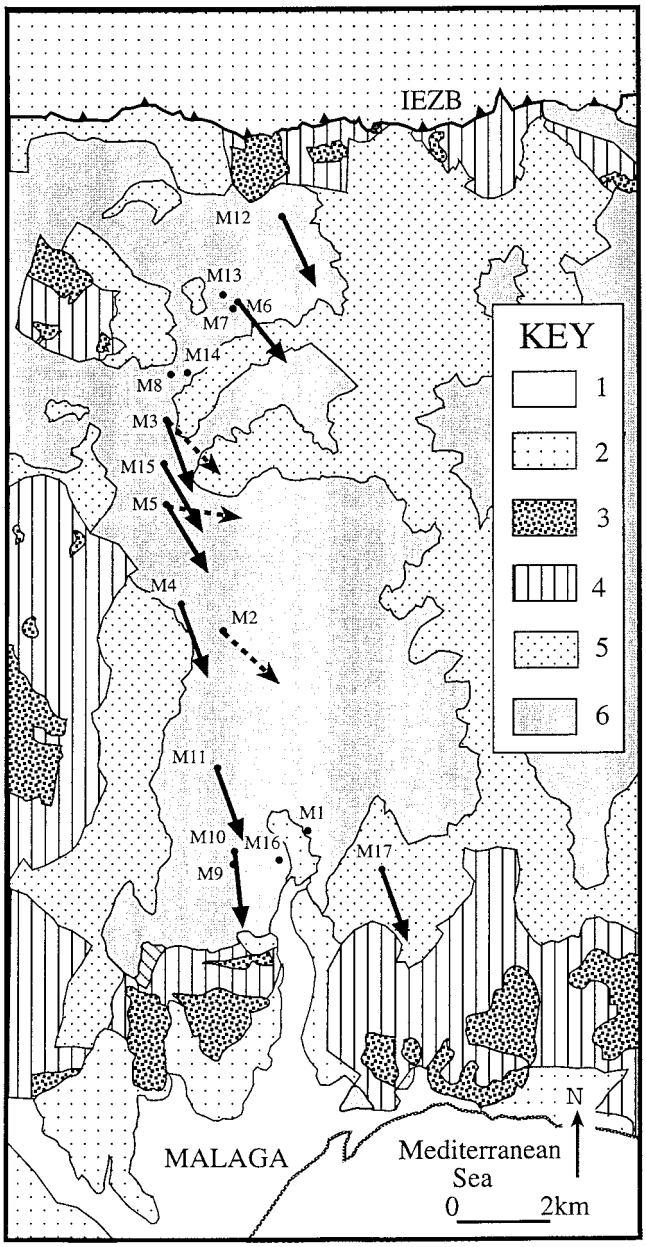


Fig. 2. Simplified geologic map of Malaguide Complex around Malaga (IGME 1978, 1979) showing palaeodeclinations of high temperature (solid arrows) after tectonic correction and intermediate temperature (dashed arrows) remanences. All remanences are shown as normal polarity. Locations correspond to Table 1. Key: 1, Quaternary; 2, Mio-Pliocene; 3, Permian to Lower Jurassic; 4, Devonian to Carboniferous; 5, Silurian to Devonian; 6, Precambrian to Ordovician.

are found in the lowest (Early Palaeozoic) unit, but some also occur in the middle (Late Palaeozoic) unit: the latter show significantly less alteration. None have been reported from the upper unit. The dykes invariably cut the ductile deformational fabrics, but are cut by the normal faults. In the area we have studied, they are for the most part sub-vertical and trend east–west. One K–Ar whole rock age of 22 Ma has been previously obtained (Torres-Roldan *et al.* 1986). The rocks have a MORB-type geochemistry, and have been attributed to shallow melting of asthenospheric mantle, following removal

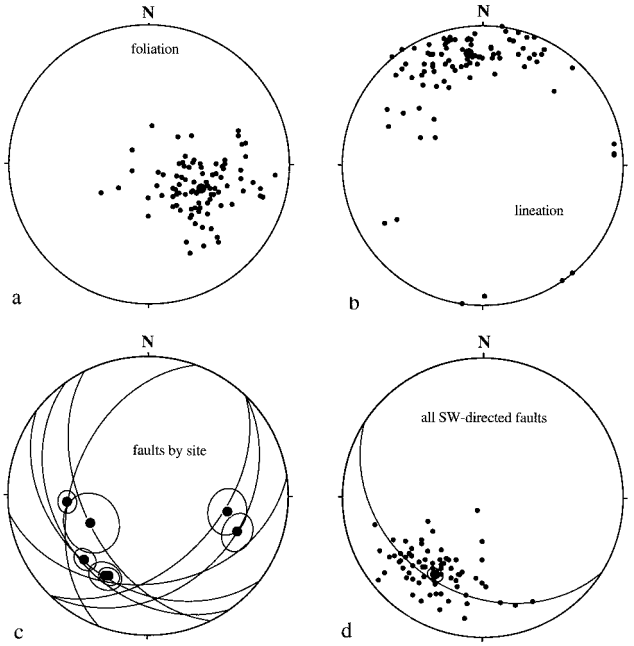


Fig. 3. Structural orientation data for the Rio Guadalmedina transect through the Malaguide complex. Heavy symbols are means. (**a**) Poles to foliation. (**b**) Ductile lineations. (**c**) Mean normal fault planes and mean slip lineations with 95% cones of confidence (Fisher statistics) from seven separate sites. (**d**) Mean fault plane and mean slip lineation for all SW-directed normal faults.

and thinning of lithospheric mantle beneath the collisional orogen in late Oligocene time (Turner *et al.* 1999).

Structure

The rocks of the lowest (Early Palaeozoic) unit in the Malaga region consist of interlayered graphitic phyllites and psammites with a strong foliation sub-parallel to lithological layering. This foliation is a finely spaced differentiated crenulation cleavage, and in thin-section an earlier slaty-type cleavage can be seen in the crenulation arcs. Both the crenulation cleavage and the earlier fabric are defined by fine-grained (10–50 μm) white mica and chlorite. In the transect that we have studied the main foliation has a moderate to steep westerly dip, with a mean value of 34° towards 295° (Fig. 3a). We found no evidence of macroscopic folding related to this foliation, but it is locally deformed by mesoscopic open folds, particularly associated with brittle normal faults.

One or more lineations lie in the foliation plane. The commonest is a fine crenulation lineation developed in the phyllites, which is likely to be the intersection of the main crenulation cleavage and the older fabric. Psammites commonly show a lineation defined by elongate deformed clastic particles and detrital mica, and by elongate pressure shadows around these particles: this is likely to be an elongation lineation, representing the finite extension direction during the deformation that produced the main fabric. The two lineations are not readily distinguishable in the field, and appear to be sub-parallel: we have therefore grouped the data together. The mean orientation of the aggregated lineation data plunges 22° towards 351° (Fig. 3b). This indicates a north–south direction of finite extension in the present-day reference frame.

The age of the deformational fabrics is not known with certainty. Because of the lack of ductile deformation in the Mesozoic rocks of the Malaguide Complex, cleavage and lineation in the underlying Palaeozoic rocks are commonly assumed to be of Variscan age (e.g., Balanyá & Garcia-Dueñas 1987). For reasons discussed later in this paper, however, we suggest that the crenulation cleavage and the dominant lineation in the lowest unit in the Malaga region may be of Alpine age, and their absence in the Mesozoic unit may reflect the presence of a significant late extensional fault separating the lowest unit from the rocks above.

The foliation in the lowest unit is locally overgrown by porphyroblasts of andalusite (now commonly altered to fine-grained phyllosilicates) and by biotite. Biotite is also associated with microscale fractures roughly normal to the main fabric, which probably served to allow ingress of fluids.

The dykes crosscut the main foliation in the lowest and middle units, and are for the most part undeformed, with a mean orientation of 80° towards 185° From a structural point of view they constitute a set of extensional fractures related to north–south extension in the present-day reference frame. The density of dykes varies: on average they have produced less than 5% extension, but locally, over north–south distances of several hundreds of metres, they have produced as much as 30% extension. Within the north–south traverse that we have studied in detail (Fig. 2) the density of dykes decreases northwards.

Ductile deformational fabrics and dykes are all cut by normal faults with widely varying trends. Most faults carry wear grooves or quartz-fibre lineations indicating the direction of motion. Displacements are for the most part not determinable, although some faults displace dykes by distances of up to 20 m. In places where the dykes are strongly disrupted by the faulting, they can be rotated into gently dipping orientations adjacent to the faults. The most common set of faults dip SW, with a mean orientation of 32° towards 212°, and a mean slip vector of 36° towards 214° (Fig. 3c and d). Subordinate east- and west-dipping faults may constitute a later conjugate set (Fig. 3c). These data suggest approximately NE–SW extension in the present-day reference frame.

Thermochronological analysis

Ar/Ar methods and data

Whole rock fragments were hand picked from crushed and 250–500 μm sieved samples. Samples were then cleaned ultrasonically in methanol and wrapped in aluminium foil prior to irradiation. They were irradiated at the RISØ reactor, Copenhagen. After irradiation, the samples were loaded into flat bottomed holes, 2 mm in diameter, 2 mm deep in an aluminium platten, each containing 20–30 such holes. Several grains of whole rock were loaded into each hole allowing sufficient space for individual grains to be fused. The plattens were loaded into an ultra-high-vacuum laser mini-chamber and placed under a heat lamp overnight to reduce atmospheric blank levels. Argon was extracted by slowly increasing the power of a focused laser until total fusion of a target grain. The laser source was a Spectron SL902TQ CW Nd/YAG laser, maximum power 17W in TEMoo mode. The beam was directed into a customized Leica Metallux 3 microscope and focused at the sample surface via the objective lens. Laser power was time-controlled via a computer-controlled external shutter. The sample was observed using a CCD camera, coaxial with the laser beam. Gases were released by laser step-heating, sequentially increasing the laser power. Sample gas was gettered using Zr/Al alloy getters and after 5 minutes, equilibrated via automated valves into a MAP 215-50 noble-gas mass spectrometer. Peaks between 35 and 41 were scanned 10 times and amounts extrapolated back to the inlet time. Blanks were analysed before and after each sample analysis and the mean of the two blanks was used to correct the sample analysis for most of the measured isotopes. Overall mean blank levels for ^{40}Ar, ^{39}Ar and ^{36}Ar are (130, 50 and 6) $\times 10^{-14}$ cm^{-3}. The resulting analyses were corrected for mass spectrometer discrimination, ^{37}Ar decay and neutron-induced interferences. The corrections used were $(^{39}Ar/^{37}Ar)Ca=0.00065$, $(^{36}Ar/^{37}Ar)Ca=0.00026$, based upon a mean from salts analysed from several irradiations and $(^{40}Ar/^{39}Ar)K=0.045$ measured in this irradiation. GA1550 biotite standard (Age=98.8 Ma, Renne *et al.* 1998) was used to monitor the neutron flux during irradiation. The samples were packed closely in a single evacuated tube and the adjacent standards yielded similar J values close to 0.00639.

Nine of the least altered dykes were analysed. Six of these nine, however, did not produce reproducible ages. Results from dyke M7A (30.2 ± 2.8 Ma) and M7B (18.5 ± 0.5 Ma) have been published in Turner *et al.* (1999). Although both these samples exhibit somewhat noisy plateaux they do show significantly different release profiles suggesting different phases dominate the release. M7A shows a saddle shaped release profile with a plateau of around 80% release in the central portion. Petrographic analysis reveals that very little, if any, original hornblende or biotite remains in M7A and therefore the 30.2 Ma date is most likely to be coming from the unaltered plagioclase. In dyke M7B the first 70% of argon release defines the plateau age of 18.5 Ma and is associated with low Ca/K ratio (*c.* 4). Given the state of alteration of these rocks, this younger age is likely to be reflecting fluid ingress and growth of new minerals, including the fine grained white mica observed in thin section, during metamorphism. As diabase clasts occur in sediments of Burdigalian age (Alonso-Chaves & Rodriguez-Vidal 1998), however, this date may represent the final stages of cooling after metamorphism.

Dyke M17, intruded into the middle (late Palaeozoic) geological unit proved to be the best site for both magnetic and geochronological analysis (Fig. 4, Table 1). Samples from this dyke show the least degree of alteration. Three steps yielded a reproducible age over 39% of the ^{39}Ar release with a weighted mean age of 23.0 ± 0.4 Ma (2σ errors). Unlike the more altered samples this sample shows very little indication of Ca/K changes through the heating schedule. Ca/K ratios are generally around 24 (though as high as 31), fall to 14 in the mid-release but rise again to 20–24 in the late release suggesting that this date comes from unaltered plagioclase. The three reproducible ages span the full variation of Ca/K, but do not meet the criteria for a true plateau age (see McDougall & Harrison 1999) and we have thus considered an isochron of all points which yields an age of 23.0 ± 2.6 Ma. Removing the first point changes the age by less than the errors and we consider this the best estimate for the final age. The 'crankshaft' style release pattern may result from a recoil artefact causing apparent age variations in the finer grained portions of this whole rock sample.

Crucially, none of these samples shows any indication of ages older than Oligocene. They do indicate, however, the dykes are likely to have been intruded over a period of time during the late Oligocene.

Palaeomagnetic analysis

Magnetic methods

Samples for magnetic analysis were obtained from 17 sites (28 dykes), 16 of which are located in the lower geological unit exposed in the

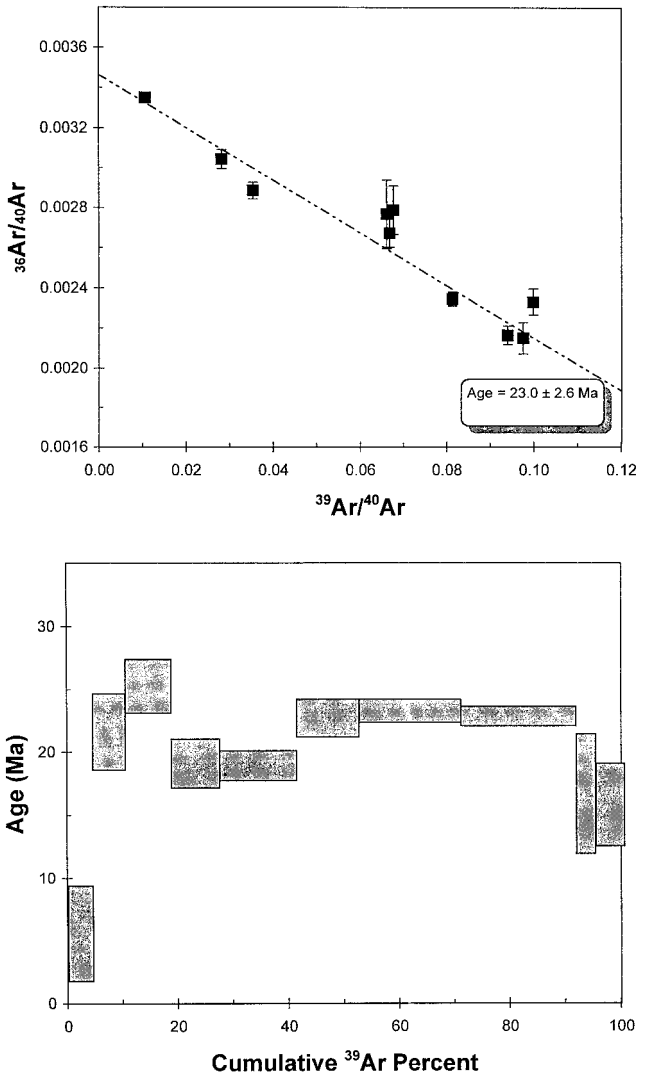


Fig. 4. Isochron (**a**) and plateau (**b**) data for dyke M17.

gorge of the river Guadalmedina, which cuts through the entire body of Malaguide rocks from the Internal/External Zone Boundary to Malaga, near the coast (Fig. 2). One site (M17) comes from the middle unit, a few kilometres NE of Malaga. At each site one to five dykes were sampled. This sampling scheme was designed to try to minimize the effects of secular variation.

Samples were obtained either with a portable petrol-powered rock drill or as oriented block samples which were taken back to the laboratory to be drilled. When cores were taken on site, five to eight cores were obtained from each dyke to allow for statistical analysis. The cores were oriented using a Brunton compass and an orienting platform which includes a sun compass for magnetically strong samples. Standard cylindrical cores of 2.5 cm diameter and 10 cm length were later cut into 2.25 cm length samples.

In the laboratory the samples were analysed to isolate the stable characteristic component of remanent magnetization (ChRM). Stepwise thermal and alternating field (AF) demagnetization procedures were used. During the thermal demagnetization procedure each sample was demagnetized with a minimum of ten steps based on a scheme designed after a detailed pilot demagnetization study. Natural remanent magnetization (NRM) measurements were made on either a two-axis cryogenic magnetometer or a Jr5A spinner magnetometer. Data obtained from the demagnetization experiments were plotted on Zijderveld vector diagrams. ChRM directions were then obtained from these plots using linear regression techniques.

On a set of representative samples, bulk susceptibility was measured between each thermal demagnetization step to monitor possible changes in magnetic mineralogy that can occur as a result of heating and dehydration.

Isothermal remanent magnetization (IRM) acquisition experiments were conducted on representative samples to peak fields between 1 and 2 T on a ASC pulse magnetizer. After saturation, smaller fields of 0.2 T and 0.05 T were imparted to orthogonal axes. The rocks were then thermally demagnetized with an ASC thermal demagnetizer.

Natural remanent magnetization

Results of thermal and AF demagnetization demonstrate that the remanence in many of the dykes is composed of one to three components (Table 2). In some dykes, however, the remanence is completely unstable and does not respond to either thermal or AF demagnetization techniques. Samples from sites M1, 7, 8, 9, 13, 14 and 16 did not carry a stable remanence.

In general, results show that the lowest temperature component usually unblocks at less than 200°C and appears to be in the direction of the present-day magnetic field. This component may also be present at the lowest coercivities (<20 mT). It is commonly unstable and poorly defined, however, and was therefore not investigated further. The intermediate temperature component unblocks between 200°C and 450°C. It has an average declination of 314° and an average inclination −48° (α_{95}=12°, k=26). The inclination is reasonable for a Miocene age, but the declination implies a very large 134° clockwise rotation. This direction correlates well with the direction obtained from the highest coercivity component, which is not completely isolated in fields of the order of 100 mT. The component of magnetization with the highest unblocking temperature is isolated between 450°C and 600°C, where the vector decays linearly to the origin of the vector plot. The declination of this vector (320°) is similar to the declination of the intermediate temperature component, but its inclination is unusually shallow (−13°). This direction correlates well with the direction of the component of NRM isolated between approximately 20 mT and 50 mT. Dykes clearly showing all three remanence components include those from sites M3 (a, c, d, e) and 5 (Fig. 5a, b).

Sites M10, M11 and M17 come from the southern part of the transect and differ from the other sites because the dykes all exhibit normal polarity. Like the more northerly sites, however, the inclination is shallower than the expected inclination of 50° for the late Oligocene. Declinations are directed towards 126°. Data from site M10 is quite noisy and shows only a single component of remanent magnetization, isolated above 350°C. Despite the large α_{95} the average direction agrees within error with the other two sites from the area. Site M11 (Fig. 5c), in contrast to site M10, shows a clearly defined normal ChRM above 350°C that records a declination of 129°. Site M17 (Fig. 5d) lies in the middle geological unit, and comes from a dyke that has been intruded into marly limestones of probable Devonian age. The NRM intensity of this dyke ($\approx$500 mA m^{-1}) is at least an order of magnitude greater than any of the other dykes in this study. This dyke has only two clearly defined components of remanent magnetization; a very low-temperature (<200°C) component and a high-temperature ChRM defined above 500°C. In many samples up to half of the original magnetization was present up to 550°. At higher temperatures (above 550°C) the magnetization falls off rapidly until 575°C, when it is <5% of its original value.

Table 1. *$^{40}Ar/^{39}Ar$ isotope data for dyke M17*

M17	$^{40}Ar/^{39}Ar$	$^{38}Ar/^{39}Ar$	$^{37}Ar/^{39}Ar$	$^{36}Ar/^{39}Ar$	^{39}Ar	Age (Ma)	±
step 1	93.852	0.203	10.28	0.314	4.51	5.7	3.8
step 2	35.435	0.063	16.03	0.108	10.37	21.6	3.1
step 3	28.260	0.045	12.94	0.082	18.68	25.2	2.1
step 4	14.977	0.036	11.52	0.040	27.33	19.1	1.9
step 5	10.018	0.040	7.96	0.023	41.14	18.9	1.2
step 6	10.254	0.045	7.00	0.022	52.32	22.7	1.5
step 7	10.640	0.043	9.10	0.023	70.71	23.3	0.9
step 8	12.300	0.060	11.95	0.029	91.35	22.8	0.8
step 9	15.109	0.065	11.63	0.042	94.87	16.7	4.7
step10	14.776	0.074	10.00	0.041	100.00	15.8	3.3

Table 2. *Remanent magnetization parameters*

Site	*N*	HT (dec/inc)	*n*	α_{95}	κ	IT (dec/inc)	*n*	α_{95}	κ
M2A*	8					335/-62	6	10.7	40.3
M2D*	8					284/-66	5	9.6	64.3
M3A	4	328/-2	4	7.1	170.6	329/-38	4	11.2	67.7
M3B†	8	330/-3	2	35.6	51.3	323/-50	5	13.6	32.6
M3C	8	326/-6	7	4.1	216.8	306/-46	7	6.6	84.9
M3D	7	336/-1	6	7.4	83.1	305/-47	6	6.5	106
M3E	8	339/0	7	3.1	388.7	315/-38	7	8.4	53
M4†	4	330/-24	2	45	31				
M5	9	318/-4	6	9.1	55.1	309/-34	4	14.5	41.2
M6	5	305/-21	5	7.5	104.7				
M10	8	127/19	6	17.6	15.5				
M11	8	129/36	7	9.7	39.7				
M12	6	328/-8	5	4.9	243.2				
M15	6	310/-15	6	14	22.2				
M17	8	123/27	7	2.6	522.1				
All		**320/-13**		9.9	22.1	**314/-48**	8	10.3	30.1

N is the number of samples demagnetized; HT (dec/inc) is the declination and inclination of the high temperature component; *n* is the number of samples used to calculate site mean; IT (dec/inc) is the declination and inclination of the intermediate temperature component; κ is the precision parameter; * indicates sites demagnetized using AF and yielding a high coercivity remanence only; † indicates sites not used in this study because of low *n*.

Dykes at site M2 were found to be unstable during thermal demagnetization, and hence were generally demagnetized using AF techniques, which isolated one or two components of remanence. The higher coercivity component has a inclination that is steeper than expected and a NW declination. In the thermal demagnetization experiments that did work there is some indication that the ChRM is, in fact, an intermediate temperature component (<400°C). A weak high temperature component, visible only in one sample (Fig. 5e) at temperatures above 450°C, appears to be oriented SE and down.

The remainder of the sites that produced interpretable data, including sites, M6, M12 and M15 (Fig. 5f), have two components of remanent magnetization. The low temperature component either shows no clear direction or present-day directions. The high temperature ChRM isolated above 400°C is oriented NW and up with an inclination that is significantly shallower than the expected values.

Isothermal remanent magnetization

Thermal demagnetization of the three axis IRM was used to characterize the magnetic minerals in these dyke rocks. IRM acquisition experiments as shown in (Fig. 6) gave curves highly suggestive of a ferromagnetic carrier. Complete saturation at fields <0.5 T effectively rules out antiferromagnetic carriers such as goethite and hematite, which saturate in much higher fields. In detail, however, there is a suggestion of an initial saturation between 0.1–0.12 mT before the final saturation at 0.3 T suggesting the presence of two low coercivity phases.

Thermal demagnetization of multicomponent IRM (Lowrie 1990) gives a much clearer indication of the magnetic minerals present in the samples. Results show that most of the magnetization resides in the intermediate (between 0.05 T and 0.2 T) and low coercivity fractions (<0.05 T) (Fig. 6a). This suggests that there is unlikely to be a significant amounts of high coercivity magnetic minerals such as goethite or haematite in these rocks. This result was corroborated by the thermal demagnetization experiments (Fig. 6b). Instead, the most likely remanence carriers are a mixture of both coarse and fine grained sulphide and magnetite with Curie temperature between 300°C and 350°C and above 500°C respectively. It was often noted, however, that in the thermal demagnetization experiments (NRM) the intermediate temperature component was not completely demagnetized at temperatures as high as

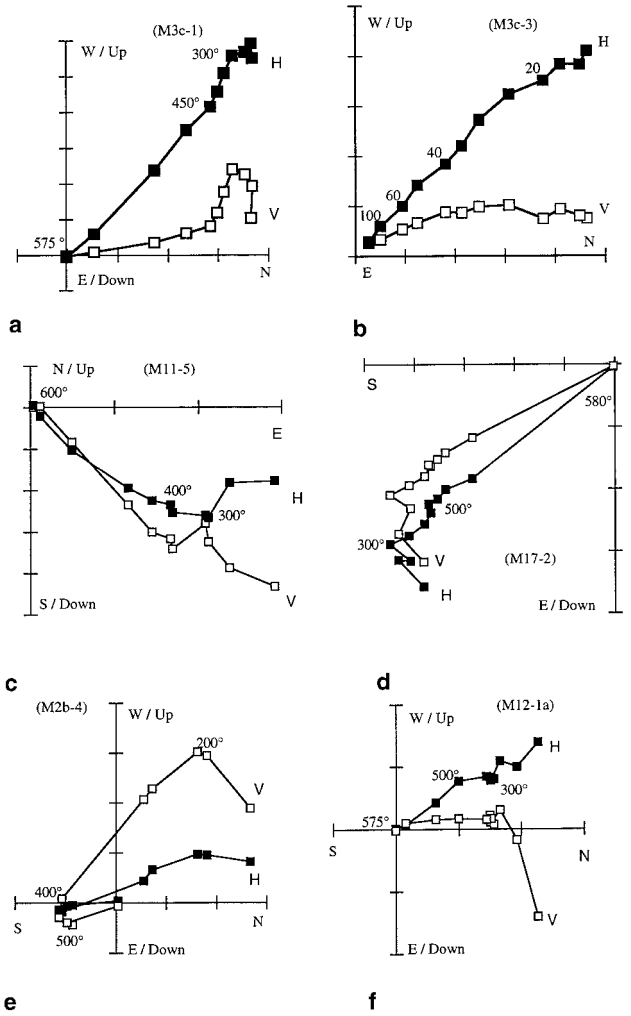


Fig. 5. Typical vector diagrams in geographic co-ordinates of samples undergoing thermal (°C) and AF (mT) demagnetization. (H, horizontal: V, vertical.)

400–450°C, well above the Curie temperature of pyrrhotite at 320°C (Dekkers 1989). One possible explanation is that the heating process is creating new magnetic phases that are inheriting their magnetization direction from their predecessors (Dunlop & Özdemir 1997). Most samples, however, do not show an increase in susceptibility until temperatures are above 500°C. An alternative explanation is that there may be some titano-magnetite/maghaemite in these samples.

Discussion

The relationship between these data sets and an explanation in terms of a history of deformation is not straightforward. The main difficulty is to explain the shallow inclination of the high-temperature component of magnetization. This component is most likely to be carried by magnetite with a remanence acquired at the time of initial emplacement of the dykes. Possible explanations for its orientation include: (a) that the region lay at low latitudes, where the magnetic field is gently inclined, at the time this magnetic component was acquired; (b) that a gently-dipping magnetic fabric was imposed on the dykes during deformation; (c) that the dykes have been tilted subsequent to magnetization.

During the late Oligocene, southern Iberia lay close to its present latitude and therefore the dykes would be expected to have an inclination of approximately 50° (Westphal *et al.* 1986). This effectively rules out the first possibility.

The second possibility is that a magnetic fabric resulting from penetrative deformation associated with the formation of the fabric in the surrounding schists may have caused the inclination in the dykes to be shallower than expected. To explore this possibility both the anisotropy of susceptibility (AMS) and the anisotropy of magnetic remanence of the dykes were measured. AMS measures the anisotropy of all magnetic minerals in the rock. These include the remanence-carrying minerals but also other paramagnetic minerals such as micas and amphiboles. Results of this work yielded an anisotropy factor of less then 1% implying that there is no preferred orientation of the magnetic minerals in the dyke rocks. In contrast to AMS, anisotropy of magnetic remanence (Jackson 1991) examines the preferred orientation of the remanence-carrying minerals only. Results of this work show that there is as much as 22% anisotropy in the orientation of the remanence carrying minerals. The shape and orientation of the anisotropy ellipise indicate that this fabric, is primarily a planar fabric (foliation, 19.6%) oriented sub-parallel to the dyke. The direction of the maximum anisotropy is steeply dipping oriented sub-parallel to the dip of the dyke (80° to 174) and the minimum anisotropy is sub-horizontal ($-10°$). This fabric is therefore unlikely to be associated with either the gently dipping fabric in the surrounding schists or the deflection of the remanence vector into the horizontal. The simplest interpretion of the observed preferred orientation is that it is a flow fabric imposed at high temperatures as the dykes were intruded into the country rock.

The third possible explanation for the shallow inclination of the high temperature component of magnetization is that the dykes were rotated about horizontal axes after magnetization. Except in the immediate vicinity of magmatic centres, dykes

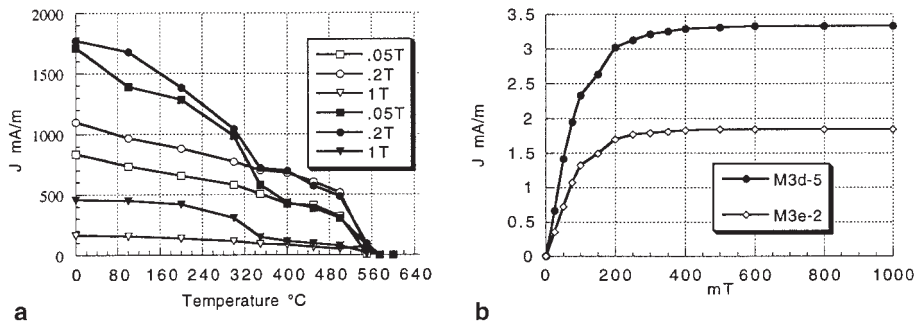


Fig. 6. (**a**) Isothermal Remanent magnetization (IRM) acquisition curves and (**b**) thermal demagnetization of three component IRM of two samples; M3d-5 (solid symbols) and M3e-2 (open symbols).

normally are intruded with steep orientations, and the uniformly steep dips of the Malaguide dykes suggests that there has been little or no change in dip since they were intruded. To explain the shallow inclination by simple tilting of the dykes about an east–west axis parallel to their trend would require a tilt towards the north of about 70° (assuming an original 50° plunge for the remanence, appropriate for a Oligocene age for the dykes). This would imply that the dykes, now essentially vertical, were intruded with a regionally consistent dip of 10°N, which is geologically unlikely. A tilt of this magnitude should also produce a detectable difference in the grade of metamorphism of the country rocks: over a north–south distance of 20 km a 70° tilt would produce a difference in the depth of burial at the present surface of 19 km, which would correspond to a pressure difference of about 5 kbar, and a likely temperature difference of 500°C or more. No such difference in the pressure or temperature of metamorphism in the country rock is detectable. Domino-style rotation on closely spaced faults could produce tilting without such large differences in metamorphic grade. The only set of structures we have found that could produce such an effect is the SW-dipping set of normal faults. The mean dip of these structures (32°) suggests that they have not experienced more than 30° of tilt, and the slip vector on these faults (36° to 214) is nearly normal to the sub-horizontal remanence, so that slip and rotation on these faults could not explain the present attitude of the remanence.

Tilting about a NNE–SSW horizontal axis could provide an acceptable solution to the problem, as it would produce little change in the orientation of the dykes, but would alter the orientation of the high-temperature component of remanence. We have somewhat arbitrarily chosen a rotation axis trending 025° (parallel to the strike of the regional foliation) to illustrate this, but we emphasize that any approximately NNE-trending rotation axis would work as well. Correction for a tilt to the WNW of 34° about an axis trending 025° gives a tilt-corrected magnetization of −43°/329°, α_{95}=10° (reversed polarity) for the high-temperature component, which is statistically indistinguishable at the 95% confidence level from the intermediate temperature component (−48°/314°, α_{95}=10°). The tilt-corrected orientation for the dykes is 87°/003°. This solution also produces less of a problem in terms of variations in the exposure level: an overall tilt in the Malaguide Complex to the WNW is reasonable in view of the exposure of unmetamorphosed Mesozoic rocks and Tertiary flysch to the west, and of the underlying metamorphosed rocks of the Alpujarride Complex to the east (Fig. 1).

An interesting consequence of the proposed tilt-correction is that it suggests that the main foliation in the rocks was subhorizontal before tilting, with a tilt-corrected stretching lineation trending 346°. The age and tectonic significance of these fabrics is uncertain, but if the foliation was originally subhorizontal it may be Alpine in age, and reflect the ductile phase of late-orogenic extension. The finite extension direction indicated by the lineation is consistent with the extension direction indicated by the dykes.

Turning this argument around and assuming that the original foliation was horizontal it is possible to correct the magnetic remanences by rotating them about a local horizontal axis defined by the local strike of the foliation (tectonic correction). In the area sampled, there is considerable variation in the orientation of the schistosity (Table 3) and consequently the amount of rotation will vary from site to site. Results of this exercise show that the mean orientation of the high temperature remanence (−46° to 337°) is not significantly changed from the direction obtained by the bulk rotation. However, the α_{95} and the precision parameter (α_{95}=5, κ=87.1) are significantly improved even using the most stringent conditions (McElhinny 1964). The angular dispersion (10°) is, in fact, somewhat less then might be predicted from models of secular variation (15°) (McFadden *et al.* 1991). It is therefore possible that because a relatively low proportion of the dykes gave useful results, we have not adequately averaged out secular variation. Given that we have sampled at least one reversal, however, it is equally possible that Oligocene secular variation is somewhat smaller then expected.

A reversal test (McFadden & McElhinny 1990) on the HT component before tectonic correction yields an angle of 26° between the mean directions of the normal and reverse observations and is therefore negative. After tectonic correction, however, this angle becomes 10°, well within the α_{95} level of confidence to give a positive Class C reversal test. The positive fold test and reversal test imply that the magnetization is more likely to have been acquired when the foliation was horizontal than in its present orientation. It is interesting to note that if we correct the data in this way the declination of the HT component after correction becomes significantly different from the IT component. This implies that there was 23°±15° clockwise rotation of this unit between the acquisition of the HT component at 30 Ma and the IT component at 18 Ma. This rotation would therefore have occurred during the extensional phase of tectonics in the Alboran domain.

Vertical axis rotation of about 134° clockwise took place after the acquisition of the intermediate temperature remanence. This is carried by pyrrhotite, and could either be a thermal remanence associated with the peak metamorphic temperatures, or it could be a chemical remanence acquired as the pyrrhotite grew above a critical size during the alteration. The fairly uniform alteration of the dykes is consistent with metamorphism under greenschist-facies conditions, and is most plausibly explained in terms of emplacment into the country rock phyllites before or during the regional metamorphism. The lack of ductile deformation in the dykes, and the fact that they cut the main foliation, however, suggests that they were emplaced late in the metamorphic history, after the ductile deformation, but possibly before the thermal peak, which produced the late andalusite and biotite porphyroblasts. This therefore leaves open the possibility that the magmatic event caused or was associated with the thermal peak.

The orientations discussed above for the foliation, lineation, dykes, and the tilt axis must all be corrected for the vertical axis rotation, giving the following values: foliation—subhorizontal; lineation trend—188°; dykes 87°/206°; tilt axis—228°.

Brittle normal faulting postdates the dykes and the metamorphic event, and therefore probably postdates acquisition of the secondary remanence. We cannot establish the relative timing of brittle faulting and vertical-axis rotation. However, in view of the coincidence between the uncorrected extension direction indicated by the normal faults (034°) and the partially corrected (Fig. 6c) extension direction indicated by both the dykes (049°) and the stretching lineation (031°), it is tempting to suggest that the rotation occurred between the ductile and brittle phases of extension. Given the existence of large clockwise vertical-axis rotations in the immediately adjacent External Zones, however, we suggest that the most plausible

Table 3. *Magnetic and structural orientation information for all sites*

Site	HT (gc)	HT (tc)	Ss	S lin	S dyke	UTM
M1					85/000 84/180	534.7/242.1
M2A			15/274	9/330 10/332	85/135 85/140	533.1/246.1
M2B			6/255	2/338	80/150 74/169	533.1/246.1
M2C			27/327		88/140	533.1/246.1
M2D					88/138 85/120	533.1/246.1
M2E			12/326	5/313	84/135 80/136	533.1/246.1
M3A	328/-2	334/-42	42/311	11/004	61/200 76/192	531.9/250.3
M3B	330/-3	337/-42	42/311	16/006	59/210	531.9/250.3
M3C	326/-6	333/-46	42/311		77/196 85/190	531.9/250.3
M3D	336/-1	344/-38	42/311		77/192	531.9/250.3
M3E	339/0	347/-36	42/311		80/175	531.9/250.3
M4	330/-24	340/-45	25/320		85/210	532.3/246.7
M5	318/-4	329/-49	48/300		80/190 74/195	532.0/248.6
M6	305/-21	322/-46	35/267		80/190 74/195	533.4/252.7
M7A			52/316		67/208 64/200	533.3/252.5
M7B			37/303	11/011	82/204 73/197	533.3/252.5
M7C			46/308		55/194 46/155	533.3/252.5
M7D			38/293	1/205	68/190 76/183	533.3/252.5
M8			12/132	6/085	90/200	532.1/251.3
M9			70/272		55/188 50/200	533.2/241.5
M10	127/19	174/61	62/274		76/170 50/200	533.3/241.7
M11	129/36	160/50	34/258		85/170	532.9/243.5
M12	328/-8	335/-42	37/308		90/184	534.1/254.2
M13			20/268		70/194	533.2/252.8
M14			44/272	7/357	90/18	532.5/251.3
M15	310/-15	330/-45	44/293	24/358	73/185	531.7/249.7
M16			33/326		66/180	534.3/254.3
M17	123/27	161/48	50/254		87/198	535.2/241.5
All	**320/-13 ± 10**	**337/-46 ± 5**	**34/295**	**22/351**	**80/185**	

HT (gc) is the mean declination/inclination of the high temperature component in geographic coordinates; HT (tc) is the mean orientation of the high temperature component in tectonic coordinates; Ss is the dip and dip direction of the average schistocity; S lin is the mean stretching lineation; S dyke is the dip and dip direction of the dykes; UTM is the location of the sampling site in UTM coordinates.

explanation for the post-IT vertical-axis rotation is that they were acquired during the thrust emplacement of the periphery of the Alboran Domain onto the Iberian margin, and therefore post-date the main period of extensional tectonics within the Alboran Domain. If this suggestion is correct, the direction of extension implied by the brittle faults must be corrected for 134° of vertical-axis rotation, giving an original orientation of 078°.

Conclusions

We suggest the following sequence of events to explain the present structural geometry and palaeomagnetic data in the Malaguide Complex around Malaga (Fig. 7).

Following Alpine contractional tectonics at an uncertain time in the Late Cretaceous to early Oligocene interval, the main foliation formed in a subhorizontal orientation in response to the start of late orogenic extension, while the lower unit was still at sufficient depth, and hence temperature, to deform by ductile processes. This conclusion must be regarded as speculative.

Intrusion of vertical NW–SE-trending dykes took place over a period of time between 23 Ma and 30 Ma (Oligocene), at which time they acquired the high-temperature magnetic remanence upon cooling through the blocking temperature of magnetite. The dating of dyke intrusion is subject to considerable uncertainty, but the original orientation of the dykes is reasonably well constrained by the data. The extension direction during dyke emplacement was 206°, subparallel to the finite extension direction of 188° during the ductile deformation.

After intrusion of the dykes, but before they acquired the intermediate-temperature remanence, the region was rotated 23 ± 15° about a vertical axis and tilted to the south by approximately 35° about a horizontal axis trending about 251°.

The peak temperature of greenschist-facies metamorphism is likely to have occurred after dyke emplacement, producing post-tectonic andalusite and biotite in the country rocks, and the greenschist-facies assemblages in the dykes.

Acquisition of the intermediate-temperature remanence in the dykes occurred during the greenschist-facies metamorphism, or during cooling at the end of the period of metamorphism and alteration.

Brittle normal faulting took place after the greenschist-facies metamorphism, probably during the early Miocene. This final phase of extension may be reflected by the cooling age of 18 Ma. The extension direction during this phase was probably east–west (079°).

Vertical-axis rotation through a clockwise angle of 134° took place after the greenschist-facies metamorphism, and probably after the end of brittle normal faulting. This may have been

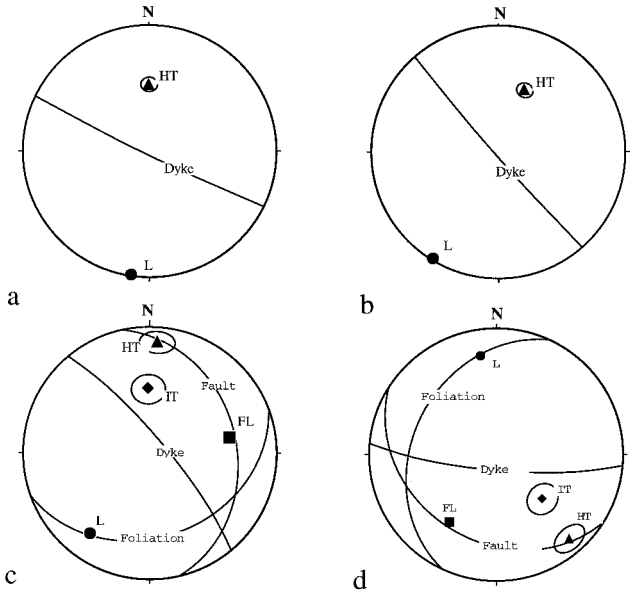


Fig. 7. Equal area stereographic projection showing orientations of magnetic and structural elements. (**a**) Data at time of intrusion. (**b**) Data rotated clockwise 23° about a vertical axis. (**c**) Data rotated 34° about a horizontal axis parallel to the strike of the foliation, acquisition of the IT and formation of brittle faults. (**d**) Data rotated 134° clockwise about a vertical axis producing observed orientations. The relative timing of rotations shown in (b) and (c) is unconstrained. Mean direction of the high temperature (HT) and intermediate temperature (IT) component of remanent magnetization are shown with α_{95} cone of confidence. L, mean stretching lineation; FL, mean fault lineation.

related to thrusting of the periphery of the Alboran Domain onto the Iberian margin.

The authors would like to thank A. Morris, P. Olivier and H. Feinberg for their careful and helpful reviews.

Note added in proof

Since this paper was accepted for publication we have obtained an Ar/Ar cooling age of 18.6 ± 1.8 Ma on fine-grained white mica from the country rock phyllites.

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Received 21 September 1999; revised typescript accepted 19 April 2000.
Scientific editing by Richard England.